Comparative Modeling Approaches for Breast Cancer Prognosis — From Gene Selection with LASSO to PCA and VAE-Based Dimensionality Reduction

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Summarize

This project evaluated multiple approaches to predicting distant recurrence-free survival (DRFS) in breast cancer using genomic data. Starting with more than 20,000 genes, univariate Cox regression identified several thousand candidates, which were further reduced to fewer than 100 genes using LASSO-Cox. We compared model performance across three input sets: (i) clinical variables only; (ii) clinical variables plus gene-expression data; and (iii) gene-expression data only, using both Cox proportional hazards models and Random Survival Forests (RSF). We also tested dimensionality-reduction methods—principal component analysis (PCA) and a variational autoencoder (VAE)—applied to the preselected genes; the resulting principal components and VAE latent variables were subsequently modeled with Cox regression.

Extensive bootstrap cross-validation was used to evaluate models by mean time-dependent AUC (iAUC) and concordance index (C-index) across iterations. Data preprocessing included Multivariate Imputation by Chained Equations (MICE) for missing clinical values, z-score normalization of gene-expression data, and strict separation of training and validation sets to prevent data leakage. Across 100 bootstrap resamples with replacement, none of the models achieved strong performance on the test set. By comparison, the PCA-based Cox model modestly outperformed the others (mean iAUC = 0.57; mean C-index = 0.56). The single best PCA–Cox model—identified on one train/test split by the highest combined score (iAUC + C-index = 1.60)—was retained as the final model for future prediction.

Introduction

Traditional survival analysis in oncology has primarily relied on clinical and pathological features such as tumor stage, grade, receptor status, and patient age. As high-throughput sequencing has become widespread and more affordable, genomic data have become central to predicting outcomes and guiding treatment decisions. The premise is straightforward: a tumor's molecular profile—specific mutations and gene-expression patterns—drives its behavior, which in turn leads to different clinical trajectories.

Transcriptomic datasets commonly used in survival studies include RNA-seq and microarrays. Standard methods in survival analysis include Kaplan–Meier curves with the log-rank test and multivariable Cox proportional hazards models. Regularized Cox models (e.g., LASSO-Cox) and Random Survival Forests are also widely applied, including for feature selection. Researchers use unsupervised clustering to identify molecular subtypes associated with distinct outcomes. In recent years, deep learning–based survival methods have gained popularity for multimodal data, complementing traditional statistical models.

In this project, we built and compared classical survival models—specifically Cox proportional hazards models and Random Survival Forests—and assessed the predictive contribution of genomic features relative to clinical variables. For genemic data, PCA was used for dimensionality reduction, and a VAE was trained to learn compact latent representations that were subsequently entered into Cox models. During exploratory analyses, patient samples were clustered by gene expression to define molecular subtypes, and survival differences across subtypes were evaluated.

Data Source

The data for this project were sourced from the internationally renowned NCBI GEO Database (Gene Expression Omnibus) under the accession number GSE7390, which includes a total of 198 breast cancer patients. All patients were lymph node-negative. Clinical variables include age, tumor grade, estrogen receptor status (ER status),and tumor size. The gene expression data were obtained using microarray technology and comprise 22,283 variables. The target variables include: Distant Metastasis-Free Survival Status(e.dmfs): A binary variable indicating whether the patient experienced disease recurrence (yes/no). Time to Distant Metastasis-Free Survival(t.dmfs): The duration from treatment to the occurrence of disease recurrence, measured in days. For patients who did not experience recurrence, the last known recurrence-free time was recorded (i.e., right-censored data).

Table 1: Subset of Gene Expression Matrix (n = 5)

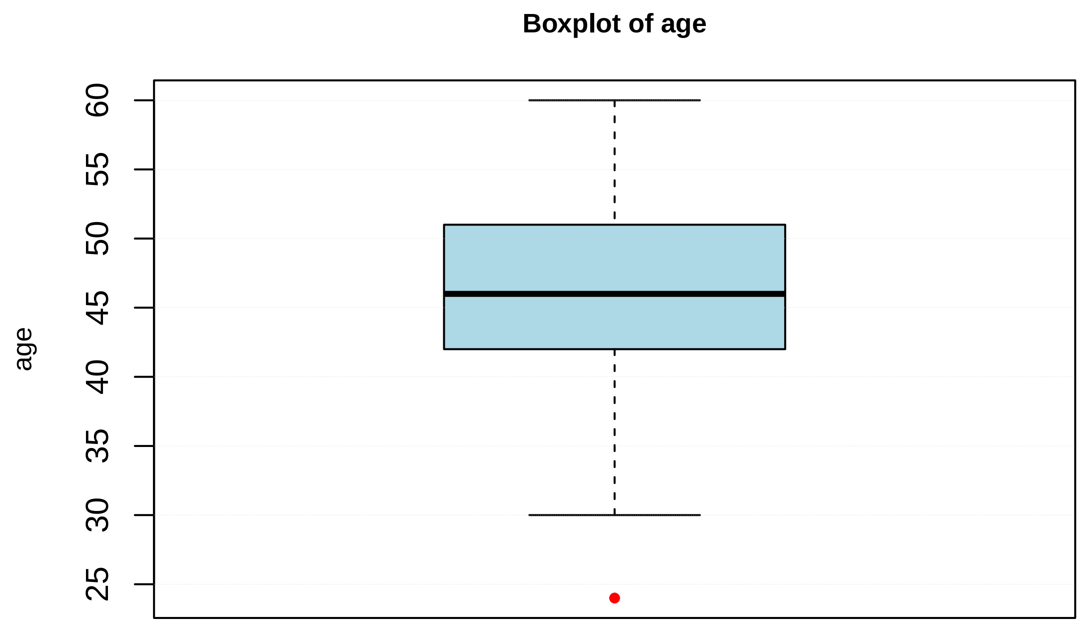
Probe_ID	**GSM177885**	**GSM177886**	**GSM177887**	**GSM177888**	**GSM177889**
1007_s_at	10.7808	11.3354	11.0281	11.8477	12.3592
1053_at	8.6742	9.3548	9.0539	9.1399	9.1967
117_at	7.7386	7.7637	6.3276	7.0326	7.8734
121_at	9.2855	9.0258	9.2344	9.6564	9.1742
1255_g_at	6.6105	4.8099	4.6220	5.5894	5.3252

Table 2: Subset of Clinical Data (n = 5)

geo_accession	**t_dmfs**	**e_dmfs**	**age**	**er**	**grade**	**size**	**hospital**
GSM177885	723	1	57	0	3	3.0	GUY
GSM177886	6591	0	57	1	3	3.0	GUY
GSM177887	524	1	48	0	3	2.5	GUY
GSM177888	6255	1	42	1	3	1.8	GUY
GSM177889	3822	1	46	1	2	3.0	GUY

Data Preprocessing and Exploratory Data Analysis (EDA)

Clean clinical data: handle missing values and outliers



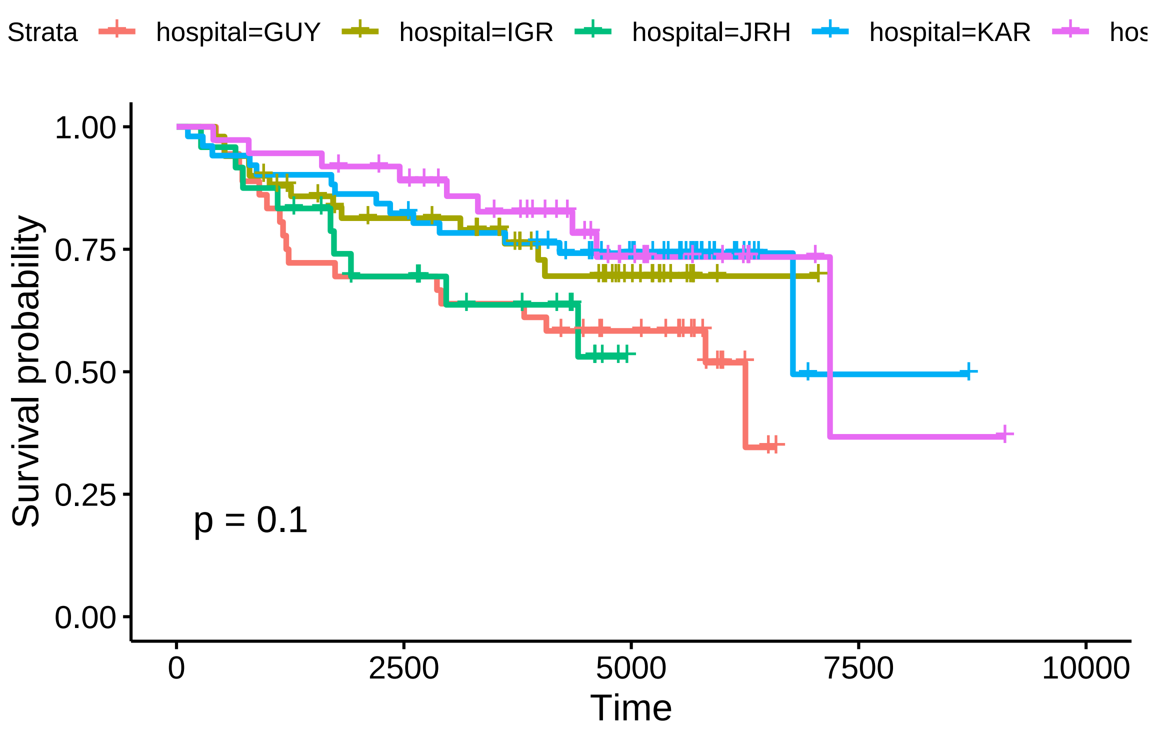
The box plot of clinical data shows no outliers.

Table 3: Table 3: NA Audit Report

Column	NA_Count	Class	Mice_Method
grade	2	factors level > 2	polyreg

The clinical dataset exhibited missingness exclusively in the grade column, with two incomplete observations. As grade represents an ordinal factor variable with three levels, multiple imputation was performed using the proportional odds logistic regression ("polr") method within the MICE framework. The final imputed values were derived through majority voting across the multiply imputed datasets.To prevent data leakage, the MICE algorithm was trained solely on the training subset. The validation set was subsequently imputed using the parameters estimated from the training data, thereby preserving the validation set's integrity for unbiased model performance assessment.

There are 5 hospitals in the clinical dataset. The Kaplan-Meier survival curves generated by ggsurvplot demonstrate partially overlapping patterns across these hospitals, and the log-rank test yields a p-value of 0.1. While this suggests some degree of heterogeneity in survival outcomes among different hospital sites, the p-value > 0.05 indicates that the observed differences are not statistically significant at the conventional alpha level of 0.05. Given the limited sample size and the lack of statistical significance, the random effects associated with hospital variation will not be considered as a primary focus in subsequent analyses.



Exploring the gene data with hierarchical cluster

The value of gene expression ranges from -2.8583908 to 16.9159279 and has multiple decimal places. This range is typical data converted by Log2. The summary table below indicates the absence of both outliers and missing values.

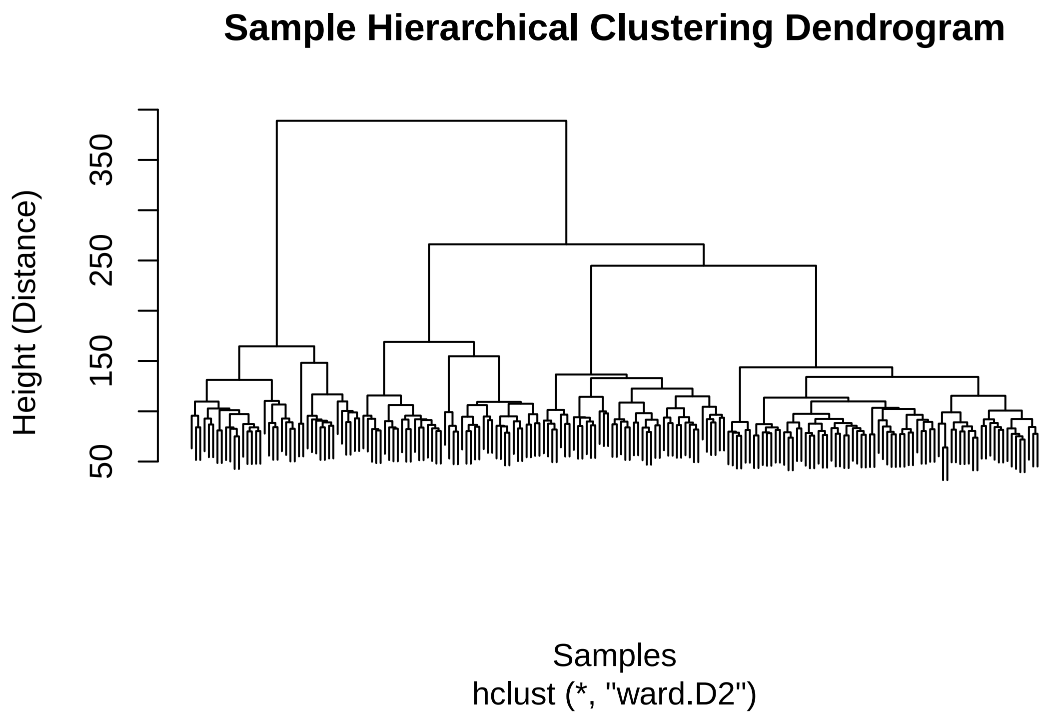
```
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.9949  5.7500  7.6778  7.6224  9.3100 16.1378

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2178  0.5802  0.8402  0.9094  1.1772  3.9008
```

Table 4: Summary of Genomic Data

Statistic	Min.	X1st.Qu.	Median	Mean	X3rd.Qu.	Max.
Mean	0.9949	5.7500	7.6778	7.6224	9.3100	16.1378
SD	0.2178	0.5802	0.8402	0.9094	1.1772	3.9008

Leveraging genomic features allows us to stratify patients into distinct molecular subgroups. We then assess whether survival curves differ meaningfully across these clusters, aiding the identification of molecular determinants that influence prognosis.

Because of computational constraints, we first performed feature preselection by taking the top 1500 most variable genes (variance ranking). We computed pairwise Euclidean distances to form a dissimilarity matrix and applied hierarchical clustering using Ward's minimum-variance linkage, which promotes compact, low–within-cluster variance groupings.

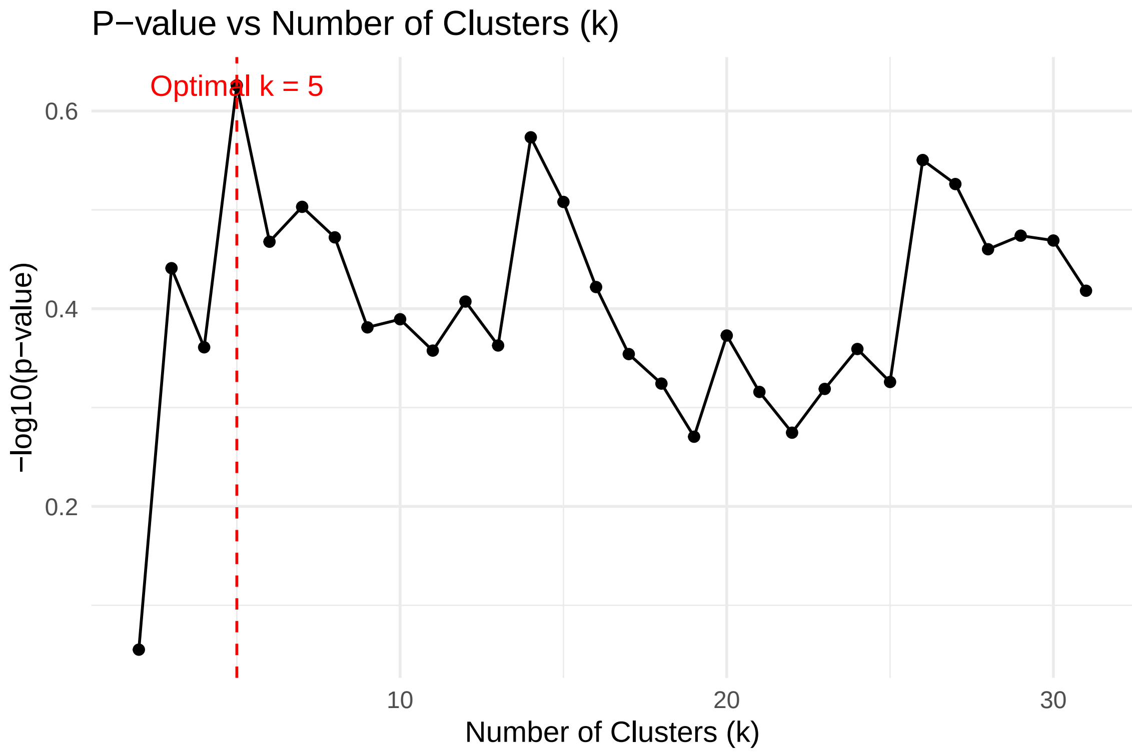


To determine the optimal clustering resolution, cluster numbers ranging from 2 to 31 were systematically evaluated. For each clustering solution, Kaplan-Meier survival analysis was conducted accompanied by log-rank testing to assess inter-cluster survival differences.

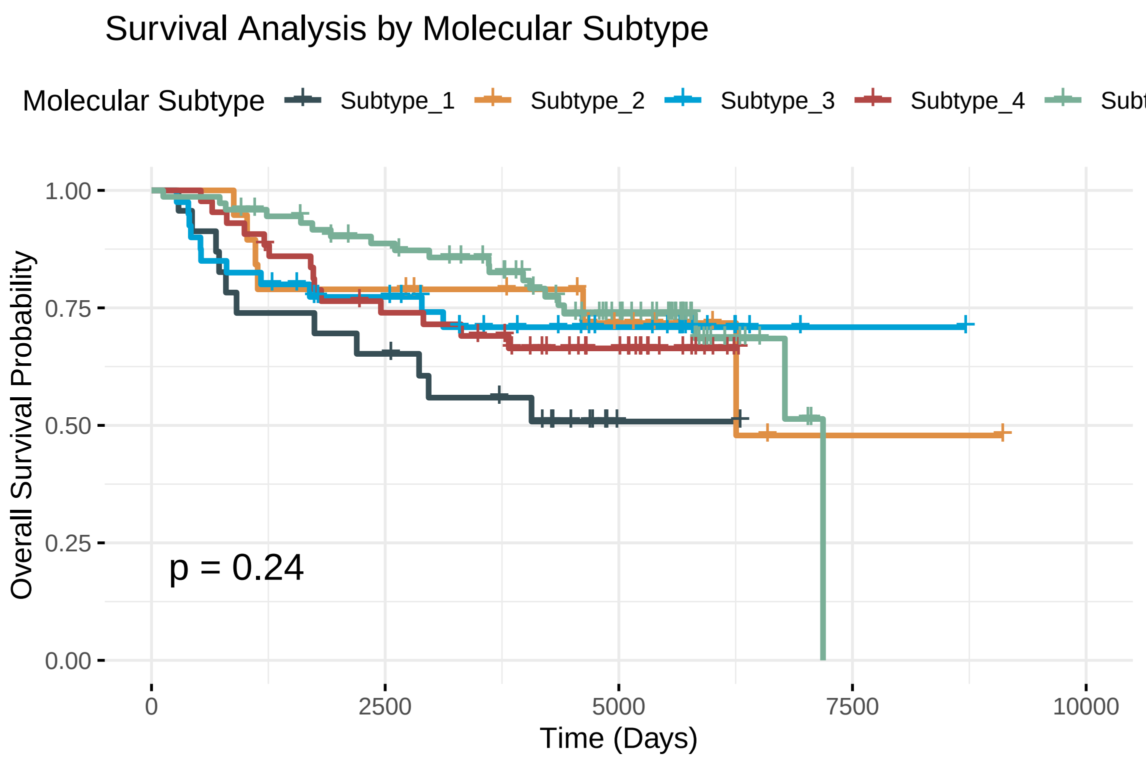
```
k = 2: p-value = 0.8808
k = 3: p-value = 0.3622
k = 4: p-value = 0.4355
k = 5: p-value = 0.2366
k = 6: p-value = 0.3406
k = 7: p-value = 0.3140
k = 8: p-value = 0.3372
k = 9: p-value = 0.4158
k = 10: p-value = 0.4080
k = 11: p-value = 0.4389
k = 12: p-value = 0.3915
k = 13: p-value = 0.4337
k = 14: p-value = 0.2671
k = 15: p-value = 0.3104
k = 16: p-value = 0.3785
k = 17: p-value = 0.4425
k = 18: p-value = 0.4740
k = 19: p-value = 0.5364
k = 20: p-value = 0.4238
k = 21: p-value = 0.4833
k = 22: p-value = 0.5314
k = 23: p-value = 0.4799
k = 24: p-value = 0.4373
k = 25: p-value = 0.4722
k = 26: p-value = 0.2816
```

```
k = 27: p-value = 0.2978
k = 28: p-value = 0.3466
k = 29: p-value = 0.3358
k = 30: p-value = 0.3396
k = 31: p-value = 0.3818
```

```
Optimal k: 5 with minimum p-value: 0.236621
```



The optimal partition 5 was selected based on the most statistically significant separation of survival curves, as determined by the minimal log-rank p-value across all tested configurations. Following the division of samples into 5 groups, the KM log-rank test yielded a p-value of 0.236621, indicating no statistically significant differences in survival outcomes among these groups.



Train Test Spliting and MAD filter on genomic data

Further processing will include: train test data splitting, clinical miss data imputation, genetic feature pre-selection, and standardization of genomic data.

```
  geo_accession t_dmfs e_dmfs age er grade size hospital
1     GSM178004   5236      0  50  1  <NA>  1.1      KAR

  geo_accession t_dmfs e_dmfs age er grade size hospital
1     GSM178004   5236      0  50  1     2  1.1      KAR
```

Trough resampling with replacement, we performed train–test splits and applied multiple imputation to the clinical covariates. For the genomic features, we applied median absolute deviation (MAD) filtering on the training set to remove genes with minimal expression variability. Genes in the top 0.75 quantile of variability were retained for downstream analysis. After preprocessing, the number of gene features was reduced from from 22283 to 16712. The validation set was restricted to the same gene subset to ensure analytical consistency and prevent data leakage.

Table 5: Training set samples for clinical data (n = 5)

	geo_accession	t_dmfs	e_dmfs	age	er	grade	size	hospital
137	GSM178021	8711	0	33	0	3	2.8	KAR
101	GSM177985	4352	0	45	0	3	4.0	JRH
37	GSM177921	7057	0	30	1	2	4.5	IGR
72	GSM177956	530	1	40	0	3	2.0	IGR
191	GSM178075	2879	0	48	0	3	3.2	RH

Table 6: Training set samples for gene expression data (n = 5)

	GSM178021	GSM177985	GSM177921	GSM177956	GSM178075
1053_at	9.1887	8.2717	8.2787	8.6067	9.3720
117_at	6.9395	7.8688	5.9665	7.5069	7.0524
1255_g_at	3.3981	4.8492	3.5869	4.7856	5.5245
1294_at	9.4102	9.3295	9.0881	8.8914	8.9788
1316_at	6.2498	7.4524	6.7756	6.9474	6.3548

COX Regression

Univariate Cox regression

We performed univariate Cox regression to screen for prognostic genes. The Cox proportional hazards model is a semi-parametric model that relates survival time to one or more predictors. Its specification is:

$$h(t|X) = h_0(t) \times \exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

where: - $h(t|X)$ is the hazard function at time t given covariates X - $h_0(t)$ is the baseline hazard function - $\exp(\beta_i X_i)$ represents the hazard ratio for each covariate

Univariate Cox proportional hazards (PH) models were fit separately for each gene. Genes showing a significant association (Wald z-test, < 0.05) were identified and compiled into a results table. The number of gene features was further reduced to 2619.

Table 7: UnivariateCox Significant Genes (n = top 5)

gene	coef	HR	SE	p.value
202859_x_at	0.4874	1.6281	0.0831	0
200760_s_at	-1.0684	0.3436	0.2072	0
214210_at	-0.8528	0.4262	0.1659	0
220106_at	0.5103	1.6658	0.0995	0
205231_s_at	-0.8915	0.4100	0.1767	0

LASSO Cox PH regression

To further refine the feature set, we then applied LASSO-penalized Cox PH regression (Least Absolute Shrinkage and Selection Operator). The L1 penalty enables simultaneous coefficient shrinkage and variable selection, yielding a parsimonious subset of prognostic genes, with the penalty strength tuned during model training. The objective function combines partial likelihood with L1 penalty:

$$\min_{\beta} \left[-\frac{1}{n}\ell(\beta) + \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

where:

- $\ell(\beta)$ = partial likelihood of the Cox model
- λ = regularization parameter controlling shrinkage
- $\sum_{j=1}^{p} |\beta_j|$ = L1 penalty term

The Lasso Cox survival modeling is to evaluate the prognostic impact of different genes/clinical variables and the relative strength of their effects. It is essential to compare effect sizes (hazard ratios) on a common scale. Therefore, Z-score normalization was applied individually to each gene and numerical clinical variable. Additional feature selection was performed using remove_high_corr_genes function with a correlation coefficient cutoff of 0.90 to address multicollinearity among predictors. To similarly prevent information leakage, the validation set was standardized utilizing the mean and standard deviation parameters calculated solely from the training dataset.

Table 8: Genomic test set (n = 5), scaled (training-set parameters)

gene	GSM177885	GSM177887	GSM177891	GSM177894	GSM177895
1255_g_at	1.4685	-0.1009	0.5835	0.7337	-0.5005
1294_at	-0.9424	-1.1126	-1.9966	-0.7195	-0.8035
1729_at	1.3576	-0.7607	0.4220	-1.6236	0.7261
200074_s_at	-0.2605	0.6378	-0.6959	0.4725	0.2196
200081_s_at	-1.5481	-0.3198	-0.7717	0.0628	0.2622

Table 9: Clinical test set (n = 5), scaled (training-set parameters)

	geo_accession	t_dmfs	e_dmfs	age	er	grade	size	hospital
1	GSM177885	723	1	1.4684	0	3	0.9500	GUY
3	GSM177887	524	1	0.2241	0	3	0.3263	GUY
7	GSM177891	5947	0	-0.3289	0	3	-0.2974	GUY
10	GSM177894	1233	1	-1.1584	1	2	0.3263	GUY
11	GSM177895	1136	1	1.7449	1	3	-0.2974	GUY

The LASSO Cox regression procedure incorporated repeated cross-validation to enhance the robustness of gene selection across multiple iterations. The resulting genes were prioritized based on selection frequency and mean coefficient magnitude, with a retention threshold set at 0.8 frequency. Regarding sample size considerations, Peduzzi et al. (1995, J Clin Epidemiol) initially proposed a conservative requirement of 10 events per variable (EPV) for Cox regression. Vittinghoff & McCulloch (2007, Am J Epidemiol) later demonstrated that 5–9 EPV may be sufficient for many analytical scenarios. Consistent with the latter guidelines and considering the cohort size limitations, an EPV threshold of 5 was implemented for additional feature selection. Since there were 69 events in the training data, the maximum number of variables allowed is 13 . Consequently, 13 genes were ultimately selected as feature variables.

However to assess whether adding gene-expression features improves the discriminative performance of Cox models over clinical covariates, we appended the genomic features directly to the clinical set without enforcing the events-per-variable (EPV) rule. The EPV constraint was likewise not re-applied after PCA-based dimensionality reduction in the next part. Owing to space limitations, a systematic evaluation of EPV-constrained specifications is left for future work.

Risk score calculation and Kplan Maier analysis between groups

The risk score for patient will be calculated as:

$$\text{risk score} = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_i x_i + \cdots + \beta_p x_p$$

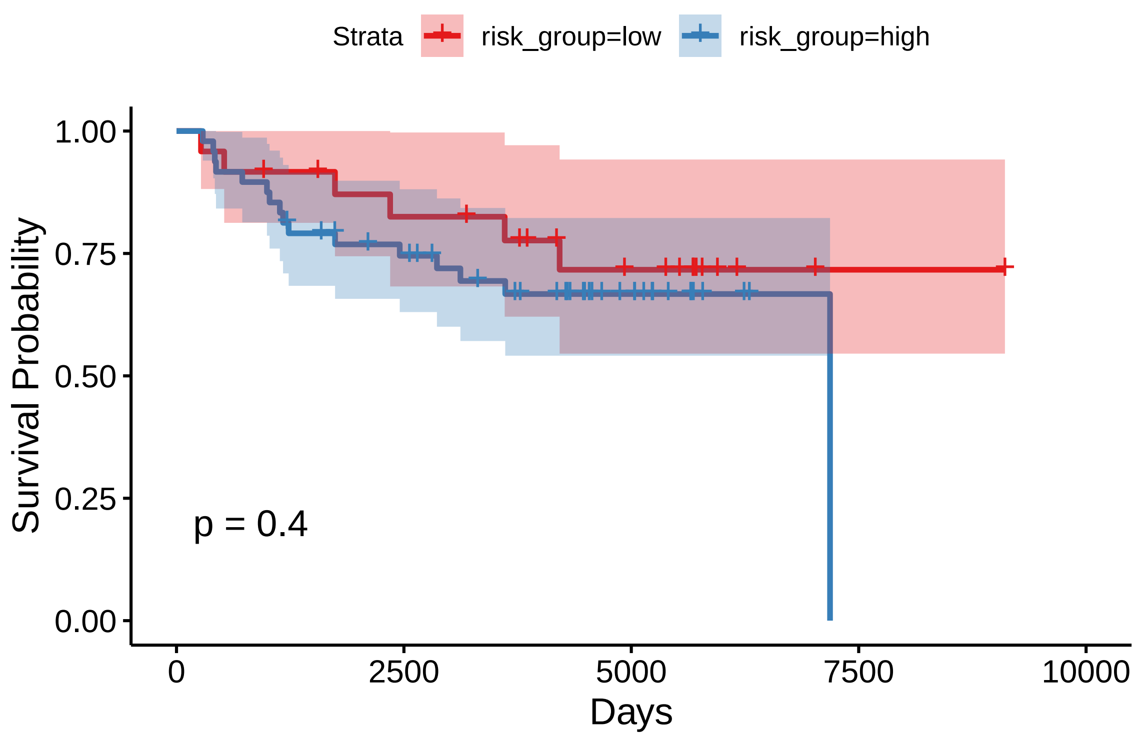
where: - β_i represents the coefficient for each variable - x_i represents the value of each predictor variable - p is the total number of variables in the model

A risk score was calculated for each patient in the training set. Subsequently, patients were stratified into high-risk and low-risk groups based on the median risk score of the training cohort. This same median value,

Table 10: Multi_Iteration_LASSO_Genes_Selection_Frequency_0.8_EPV_5

	gene	freq	mean_coef
202859_x_at	202859_x_at	1	0.5981
220106_at	220106_at	1	0.3690
216168_at	216168_at	1	0.3173
204531_s_at	204531_s_at	1	0.2824
217263_x_at	217263_x_at	1	-0.1861
200760_s_at	200760_s_at	1	-0.1856
215502_at	215502_at	1	-0.1679
214013_s_at	214013_s_at	1	-0.1378
218054_s_at	218054_s_at	1	-0.1208
215969_at	215969_at	1	0.1163
215348_at	215348_at	1	0.1122
214210_at	214210_at	1	-0.1074
201815_s_at	201815_s_at	1	-0.1050

derived from the training set, was then applied as the cutoff to classify patients in the validation set into high- and low-risk groups following the calculation of their risk scores.



The Kaplan–Meier survival curves show substantial overlap between the two groups, and the log-rank test confirms no statistically significant difference in survival (p = 0.399324).

Safeguards Against Separation in Cox Models

Cox models can exhibit (quasi-)complete separation, causing coefficients to diverge and the concordance index (C-index) to appear spuriously high. To screen out such pathological fits, we enforce two checks: if the model's C-index exceeds min_concord (e.g., 0.97) or any coefficient's absolute value exceeds coef_max (e.g., 10 on the log-hazard scale), the model is rejected. The following shows the model's validation results from one random train–test split. Next, we conduct repeated cross-validation to evaluate overall performance.

```
Cox model summary:
Call:
coxph(formula = formula, data = df, x = TRUE, y = TRUE)

  n= 198, number of events= 69

                  coef exp(coef) se(coef)      z Pr(>|z|)
age             0.1362    1.1459   0.1630  0.835  0.40361
er1            -0.2095    0.8110   0.3943 -0.531  0.59514
grade2          0.8709    2.3890   0.4098  2.125  0.03358 *
grade3          0.4722    1.6035   0.5371  0.879  0.37936
size           -0.2921    0.7467   0.1809 -1.615  0.10637
ge_202859_x_at  1.3924    4.0243   0.2385  5.838 5.27e-09 ***
ge_220106_at    1.2306    3.4234   0.2112  5.828 5.61e-09 ***
ge_216168_at    1.0747    2.9291   0.2014  5.337 9.46e-08 ***
ge_204531_s_at  1.2796    3.5954   0.2208  5.796 6.81e-09 ***
ge_217263_x_at -1.0729    0.3420   0.2109 -5.088 3.62e-07 ***
ge_200760_s_at -0.2386    0.7877   0.1790 -1.333  0.18269
ge_215502_at   -0.6351    0.5299   0.2313 -2.746  0.00604 **
ge_214013_s_at -0.5531    0.5752   0.1917 -2.885  0.00392 **
ge_218054_s_at -0.6223    0.5367   0.1968 -3.162  0.00157 **
ge_215969_at    0.8210    2.2728   0.1898  4.326 1.52e-05 ***
ge_215348_at    0.5922    1.8080   0.2058  2.877  0.00401 **
ge_214210_at   -0.8360    0.4334   0.2045 -4.088 4.36e-05 ***
ge_201815_s_at -0.2712    0.7625   0.1554 -1.745  0.08091 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

               exp(coef) exp(-coef) lower .95 upper .95
age               1.1459     0.8727    0.8324    1.5773
er1               0.8110     1.2331    0.3744    1.7564
grade2            2.3890     0.4186    1.0700    5.3337
grade3            1.6035     0.6236    0.5596    4.5950
size              0.7467     1.3393    0.5237    1.0645
ge_202859_x_at    4.0243     0.2485    2.5217    6.4222
ge_220106_at      3.4234     0.2921    2.2632    5.1784
ge_216168_at      2.9291     0.3414    1.9739    4.3466
ge_204531_s_at    3.5954     0.2781    2.3324    5.5422
ge_217263_x_at    0.3420     2.9239    0.2262    0.5170
ge_200760_s_at    0.7877     1.2694    0.5546    1.1189
ge_215502_at      0.5299     1.8872    0.3367    0.8338
ge_214013_s_at    0.5752     1.7386    0.3950    0.8376
ge_218054_s_at    0.5367     1.8632    0.3650    0.7893
ge_215969_at      2.2728     0.4400    1.5668    3.2970
ge_215348_at      1.8080     0.5531    1.2078    2.7065
ge_214210_at      0.4334     2.3072    0.2903    0.6472
ge_201815_s_at    0.7625     1.3116    0.5623    1.0339

Concordance= 0.949  (se = 0.008 )
Likelihood ratio test= 266.8  on 18 df,   p=<2e-16
Wald test            = 107.6  on 18 df,   p=9e-15
```

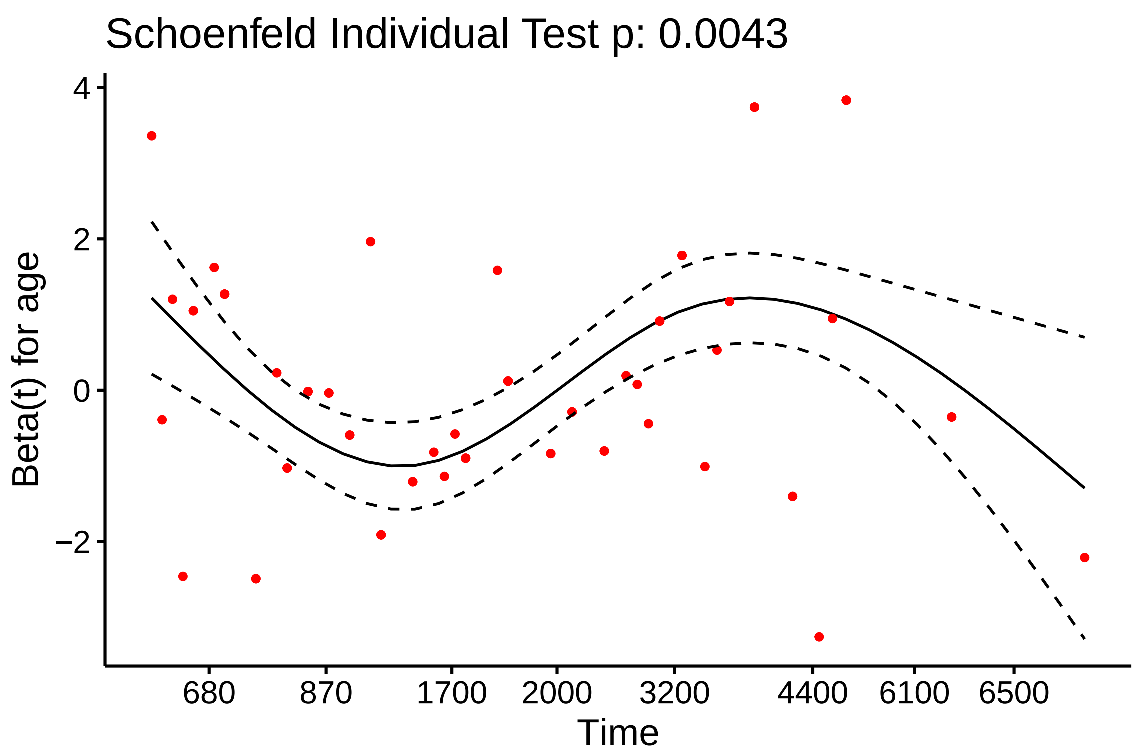
```
Score (logrank) test = 250.3  on 18 df,   p=<2e-16

Proportional hazards (PH) assumption test:
                 chisq df       p
age              8.138  1 0.00434
er               2.779  1 0.09552
grade           23.455  2 8.1e-06
size             5.057  1 0.02453
ge_202859_x_at   0.262  1 0.60883
ge_220106_at     0.442  1 0.50635
ge_216168_at     0.437  1 0.50878
ge_204531_s_at   1.237  1 0.26598
ge_217263_x_at  14.955  1 0.00011
ge_200760_s_at   4.696  1 0.03024
ge_215502_at    13.269  1 0.00027
ge_214013_s_at  11.434  1 0.00072
ge_218054_s_at   0.621  1 0.43062
ge_215969_at    10.970  1 0.00093
ge_215348_at     5.814  1 0.01590
ge_214210_at     1.441  1 0.22997
ge_201815_s_at   0.317  1 0.57367
GLOBAL          96.265 18 1.1e-12
```

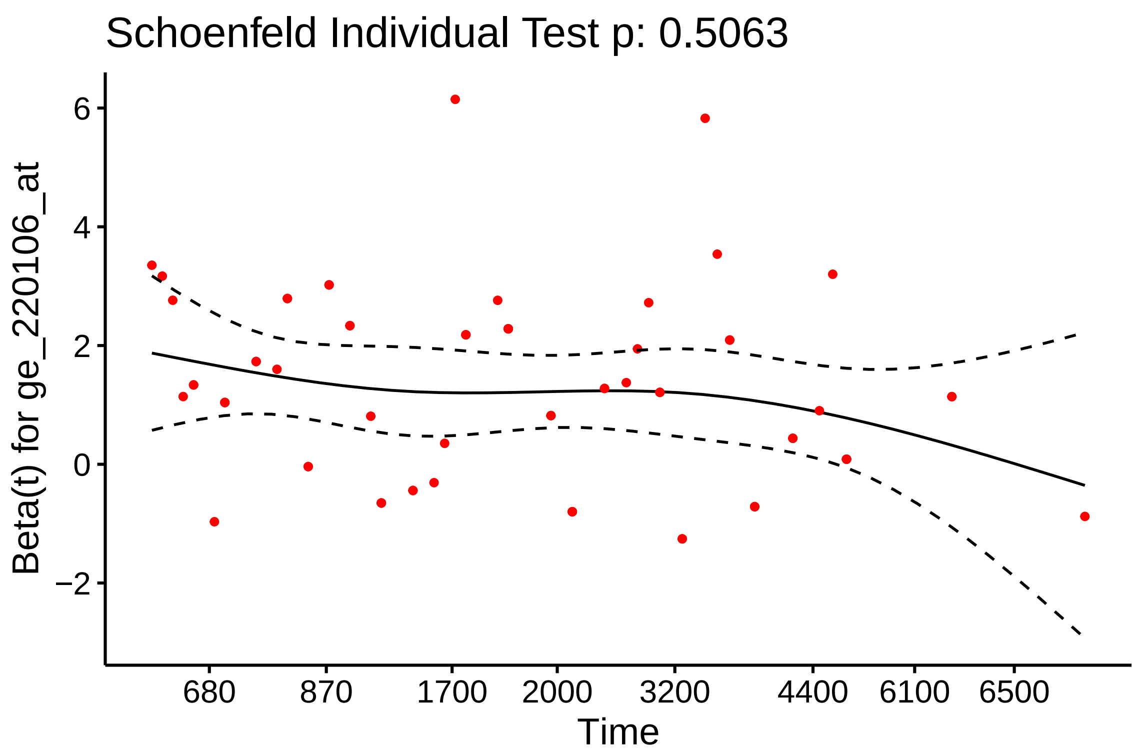
Table 11: Full Cox Model Analysis Summary

	variable	**coef**	**zscore**	**pvalue**
age	age	0.1362	0.8352	0.4036
er1	er1	-0.2095	-0.5314	0.5951
grade2	grade2	0.8709	2.1251	0.0336
grade3	grade3	0.4722	0.8791	0.3794
size	size	-0.2921	-1.6147	0.1064
ge__202859__x__at	ge__202859__x__at	1.3924	5.8384	0.0000
ge__220106__at	ge__220106__at	1.2306	5.8280	0.0000
ge__216168__at	ge__216168__at	1.0747	5.3368	0.0000
ge__204531__s__at	ge__204531__s__at	1.2796	5.7956	0.0000
ge__217263__x__at	ge__217263__x__at	-1.0729	-5.0881	0.0000
ge__200760__s__at	ge__200760__s__at	-0.2386	-1.3325	0.1827
ge__215502__at	ge__215502__at	-0.6351	-2.7458	0.0060
ge__214013__s__at	ge__214013__s__at	-0.5531	-2.8845	0.0039
ge__218054__s__at	ge__218054__s__at	-0.6223	-3.1623	0.0016
ge__215969__at	ge__215969__at	0.8210	4.3256	0.0000
ge__215348__at	ge__215348__at	0.5922	2.8774	0.0040
ge__214210__at	ge__214210__at	-0.8360	-4.0875	0.0000
ge__201815__s__at	ge__201815__s__at	-0.2712	-1.7454	0.0809

Table 12: Full Cox Model Proportional Hazards Test

	chisq	df	p
age	8.1376	1	0.0043
er	2.7789	1	0.0955
grade	23.4547	2	0.0000
size	5.0571	1	0.0245
ge__202859__x__at	0.2619	1	0.6088
ge__220106__at	0.4416	1	0.5063
ge__216168__at	0.4366	1	0.5088
ge__204531__s__at	1.2373	1	0.2660
ge__217263__x__at	14.9550	1	0.0001
ge__200760__s__at	4.6958	1	0.0302
ge__215502__at	13.2689	1	0.0003
ge__214013__s__at	11.4337	1	0.0007
ge__218054__s__at	0.6212	1	0.4306
ge__215969__at	10.9696	1	0.0009
ge__215348__at	5.8137	1	0.0159
ge__214210__at	1.4410	1	0.2300
ge__201815__s__at	0.3166	1	0.5737
GLOBAL	96.2649	18	0.0000






```
Cox model summary:
Call:
coxph(formula = formula, data = df, x = TRUE, y = TRUE)

  n= 198, number of events= 69

           coef exp(coef) se(coef)      z Pr(>|z|)
age     0.03636   1.03703  0.12081  0.301    0.763
er1    -0.34581   0.70764  0.30476 -1.135    0.256
grade2 -0.13227   0.87610  0.34021 -0.389    0.697
grade3 -0.49921   0.60701  0.41798 -1.194    0.232
size    0.06948   1.07195  0.12083  0.575    0.565

       exp(coef) exp(-coef) lower .95 upper .95
age       1.0370     0.9643    0.8184     1.314
er1       0.7076     1.4131    0.3894     1.286
grade2    0.8761     1.1414    0.4497     1.707
grade3    0.6070     1.6474    0.2676     1.377
size      1.0720     0.9329    0.8459     1.358

Concordance= 0.524  (se = 0.039 )
Likelihood ratio test= 2.38  on 5 df,   p=0.8
Wald test            = 2.32  on 5 df,   p=0.8
Score (logrank) test = 2.32  on 5 df,   p=0.8

Proportional hazards (PH) assumption test:
        chisq df       p
age     9.198  1 0.00242
er      0.414  1 0.51999
```

```
grade  11.997  2 0.00248
size    0.977  1 0.32291
GLOBAL 23.988  5 0.00022
```

Table 13: Clinical Cox Model Analysis Summary

	variable	coef	zscore	pvalue
age	age	0.0364	0.3010	0.7634
er1	er1	-0.3458	-1.1347	0.2565
grade2	grade2	-0.1323	-0.3888	0.6974
grade3	grade3	-0.4992	-1.1943	0.2323
size	size	0.0695	0.5750	0.5653

Table 14: Clinical Cox Model Proportional Hazards Test

	chisq	df	p
age	9.1977	1	0.0024
er	0.4139	1	0.5200
grade	11.9968	2	0.0025
size	0.9771	1	0.3229
GLOBAL	23.9878	5	0.0002

Random Survival Forest model

Following the identification of significant genes, subsequent modeling will employ two distinct algorithms: a Cox proportional hazards model and a Random Survival Forest model.

Random Survival Forest (RSF) is an extension of the Random Forest method to the survival analysis domain.It consists of many survival trees, built using bootstrap samples and random feature selection at each split. Splitting rule: Instead of maximizing information gain (e.g., Gini index or entropy in classification), survival trees use the log-rank test to find the feature and split point that best separate survival outcomes. For standard random forests, the leaf output is class label (classification) or numeric value (regression). In RSF, each terminal node (leaf) stores an estimated cumulative hazard function (CHF) (H(t)). For a new sample, each tree returns the CHF of the leaf the sample falls into, and the ensemble prediction is the average CHF across trees $H_{\text{avg}}(t)$ and convert to a survival function:

$$S(t) = \exp(-H_{\text{avg}}(t))$$

.

RSF outputs a survival curve, which gives the estimated probability of surviving beyond any time point (t). Besides individual survival prediction, RSF also supports variable importance analysis. With the library(randomForestSRC) `rfsrc(Surv(t_dmfs, e_dmfs) ~ ., data = train_data, ntree = ntree)`can be used to build the RSF model.

This analysis utilized a random survival forest comprising 1,000 trees, with model training conducted through k-fold cross-validation, followed by selection of the optimal model for testing evaluation.Variable importance was determined using the full dataset.

```
Fold 1
Fold 2
Fold 3
Fold 4
```

```
Fold 5
Best Fold: 4
Best C-index: 0.9616
[1] "Variable Importance(retrained on full data):"
          age             er          grade           size ge_202859_x_at
  0.0255159821   0.0008710081   0.0009558587   0.0067722541   0.1778008797
  ge_220106_at   ge_216168_at ge_204531_s_at ge_217263_x_at ge_200760_s_at
  0.1045650272   0.0529578308   0.1432007703   0.0895792947   0.0951136786
  ge_215502_at ge_214013_s_at ge_218054_s_at   ge_215969_at   ge_215348_at
  0.0918682582   0.0253199509   0.0558481400   0.0982419248   0.0289682258
  ge_214210_at ge_201815_s_at
  0.0667379623   0.0592925421
```

Table 15: Variable Importance from Random Survival Forest model

	x
age	0.0255
er	0.0009
grade	0.0010
size	0.0068
ge_202859_x_at	0.1778
ge_220106_at	0.1046
ge_216168_at	0.0530
ge_204531_s_at	0.1432
ge_217263_x_at	0.0896
ge_200760_s_at	0.0951
ge_215502_at	0.0919
ge_214013_s_at	0.0253
ge_218054_s_at	0.0558
ge_215969_at	0.0982
ge_215348_at	0.0290
ge_214210_at	0.0667
ge_201815_s_at	0.0593

Model Discrimination Assessment Metrics

C-index (Concordance Index)

In this study, the discriminatory power of the survival model was assessed using the **average C-index** (Concordance Index) and **average iAUC** (Integrated Area Under the Curve) obtained through repeated cross-validation. The C-index is calculated by randomly selecting a pair of individuals (i, j). The pair is considered **concordant** if: - Individual i has an actual event time shorter than individual j, **and** - The model assigns a higher risk score (or shorter predicted survival time) to individual i. The C-index represents the proportion of concordant pairs among all comparable pairs:

$$C = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}}$$

Interpretation of Values: - $C = 0.5 \rightarrow$ Prediction is no better than random guessing - $C = 1.0 \rightarrow$ Perfect discriminatory ability - $0.7 \leq C < 0.8 \rightarrow$ Acceptable discrimination - $0.8 \leq C < 0.9 \rightarrow$ Excellent discrimination - $C \geq 0.9 \rightarrow$ Outstanding discrimination The C-index primarily assesses risk ordering but does not account for time-dependent dynamics in model performance.

iAUC (Integrated Area Under the Curve)

The iAUC is the integrated form of time-dependent AUC. It measures a model's discriminatory ability at different time points and provides an overall summary score through integration (weighted averaging) over time. The calculation proceeds in two main steps: i. **Compute time-dependent AUC(t)**: At a specific time point t, define two groups:

```
    -   The **"event group"**: individuals who experienced the event by time $t$.
    -   The **"non-event/survival group"**: individuals who remain event-free beyond time $t$ (including

-   Calculate the standard AUC (Area Under the ROC Curve) for classifying these two groups at time $t$
```

ii. **Integrate over all observed time points**: The overall iAUC is obtained by integrating the time-dependent AUC(t) function over the entire follow-up period $[0, \tau]$:

```
$$
\text{iAUC} = \int_{0}^{\tau} \text{AUC}(t) \cdot w(t)  \, dt
$$

Where:
-   $\tau$ is the maximum observation time.
-   $w(t)$ is a weight function, often based on the **distribution of event times** or the **size of th
```

Interpretation of Values: - iAUC = 0.5 → Discriminatory power equivalent to **random guessing**. - iAUC = 1.0 → **Perfect discriminatory ability** across all time points. - Values between 0.5 and 1.0 indicate the degree of predictive accuracy, with higher values representing better performance.

Both the C-index and iAUC provide valuable insights into model performance: The C-index offers a straightforward measure of a model's overall ability to rank patients by risk. The iAUC provides a more comprehensive view of time-dependent predictive performance. The combination of these metrics through repeated cross-validation ensures a robust assessment of the survival model's discriminatory power. The table below shows the C-index and iAUC of the COX and RSF models, which were built using both genetic and clinical data, along with the performance of models using clinical data alone, all evaluated on the test set

```
Time-weighted average AUC (iAUC): 0.7452
Concordance Index (C-index): 0.7257
```

```
Time-weighted average AUC (iAUC): 0.7623
Concordance Index (C-index): 0.7537
```

```
Time-weighted average AUC (iAUC): 0.7772
Concordance Index (C-index): 0.7081
```

Table 16: Model Performance Comparison

Model	iAUC	C_Index	Sum_iAUC_Cindex
COX Clinical-Genomic	0.7451841	0.7257	1.470884
RSF Clinical-Genomic	0.7772074	0.7081	1.485307
COX Clinical-only	0.7622508	0.7537	1.515951

Botstrap cross validation using three separate feature sets

To compare the performances of the Cox proportional hazards model and the Random Survival Forest model, Each algorithm were trained and evaluated using three separate feature sets: an exclusive clinical variable set, an exclusive gene expression variable set, and a combined set integrating both clinical and genomic features. Through bootstrap cross validations with replacement, we calculate the everage of C_index and iAUC.

Table 17: 3 Model Performance Comparision from 100 Bootstrap Iterations

Model_Type	Mean_Cox_iAUC	Mean_Cox_Cindex	Mean_RSF_iAUC	Mean_RSF_Cindex	Cox_Sum	RSF_Sum
Clinical-only	0.5544709	0.5237260	0.5164750	0.5001670	1.078197	1.016642
Genomic-only	0.5212806	0.5161470	0.5199415	0.5076450	1.037428	1.027586
Clinical-Genomic Integrated	0.5410127	0.5391948	0.5220367	0.5039948	1.080207	1.026031

Across 100 bootstrap resamples, the Clinical-Genomic Integrated Cox model exhibited a slight performance advantage over the alternative models; however, its mean iAUC and mean C-index were only 0.5410127 and 0.5391948, respectively, indicating limited discriminative ability.

##PCA Cox: Cox model built on Principal Component Analysis (PCA) results

PCA is an unsupervised, linear method that rotates the data to new, orthogonal axes (principal components) that capture the largest possible variance first. Each sample gets scores on these components; each gene has loadings that show how strongly it contributes. PCA is typically computed by an eigen-decomposition of the covariance (or correlation) matrix, or via singular value decomposition (SVD).

PCA reduces dimensionality, mitigates multicollinearity, and can denoise high-dimensional data. Keeping only the top components often preserves most of the signal while discarding noise, which can improve model stability and computation. In gene expression studies, PCA can compress ~20,000 genes into a few dozen components that retain most variation, yielding more stable estimates and faster computation in downstream analyses.

In this project, we reduced the dimensionality of gene-expression data via PCA fitted on the training set and used the resulting principal component (PC) scores as predictors in Cox proportional hazards (PH) models. The validation set was projected using the training-set centering, scaling, and rotation (loadings). Model performance was summarized over 100 bootstrap resamples. When selecting the number of PCs, we enforced an events-per-variable (EPV) rule of 3. The goal is to ensure that the retained principal components explain at least 50% of the variance in the original data.

Table 18: Cox model built on PCA CV result (n_runs = 100)

Metric	Mean	SD
C-index	0.5713	0.1127
CoxMe C-index	0.4217	0.0718
iAUC	0.5624	0.1211
Total Score	1.1336	0.2240

Exploratory analyses suggested between-hospital heterogeneity in survival ($p = 0.10$). Motivated by this, we also fit a Cox mixed-effects model (coxme) with a hospital-level random effect, but it did not improve validation performance. The mean C-index performance fell below random chance levels. This sub-random performance is presumably attributable to severe sample insufficiency after incorporating random effects into the model.

Table 19: 4 Model Performance Comparision

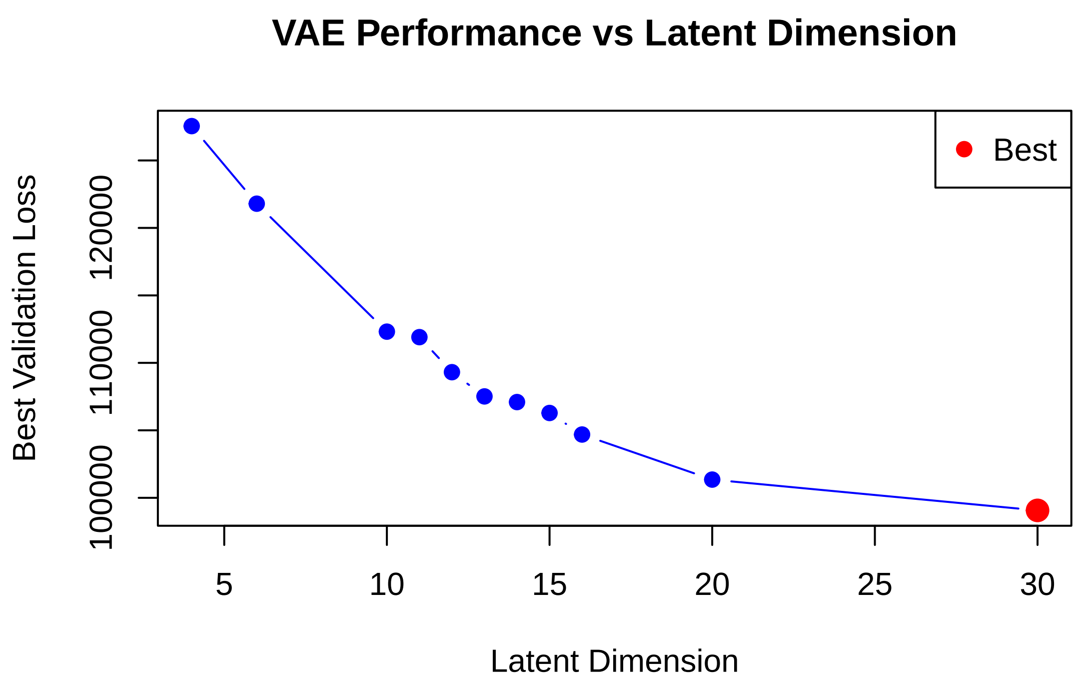
Model_Type	Mean_Cox_iAUC	Mean_Cox_Cindex	Mean_RSF_iAUC	Mean_RSF_Cindex	Cox_Sum	RSF_Sum
Clinical-only	0.5544709	0.5237260	0.5164750	0.5001670	1.078197	1.016642
Genomic-only	0.5212806	0.5161470	0.5199415	0.5076450	1.037428	1.027586
Clinical-Genomic Integrated	0.5410127	0.5391948	0.5220367	0.5039948	1.080207	1.026031
PCA-Genomic-only	0.5712670	0.5623551	NA	NA	1.133622	NA

It can be observed that after 100 iterations of bootstrapped cross-validation, the Cox model built on PCA principal components achieved mean C-index and iAUC values of 0.5623551 and 0.571267, respectively. Again this indicates suboptimal predictive power but they are slightly better than other models.

VAE Cox: Cox model built on VAE latent representation

A Variational Autoencoder (VAE) is a type of deep generative model that learns to represent high-dimensional data, such as gene expression profiles, in a lower-dimensional latent space.In this project this method was also explored, with the resulting latent variables incorporated into Cox regression modeling.

VAE consists of two parts: an encoder, which compresses the input data into a set of latent variables, and a decoder, which reconstructs the data from these variables. Unlike a standard autoencoder, a VAE assumes a probabilistic distribution for the latent variables (typically Gaussian), allowing it to generate new data samples by sampling from this distribution. This property makes VAEs useful not only for dimensionality reduction but also for capturing complex patterns in the data and creating smooth latent representations that can be used in downstream analyses, such as survival modeling.



In this project, VAE models were first trained on the training dataset across multiple latent dimensions 4, 6, 10, 11, 12, 13, 14, 15, 16, 20, 30, and the reconstruction loss + Kullback–Leibler (KL) divergence was evaluated on an independent test set. The latent dimension 13 yielding the lowest loss was then selected as the basis for subsequent Cox regression modeling.

In the next step a VAE model was first constructed using the training data, from which the latent representations of training data were extracted. A Cox model based solely on these latent variables was then built. The validation data were subsequently transformed into latent representations with the VAE model and used to evaluate the Cox model.Fifty iterations of cross-validation with replacement were conducted.

It is worth noting that, to reduce computational load in the VAE pipeline, we preselected genes prior to training by (i) applying median absolute deviation (MAD) filtering to remove low-variability genes, (ii) conducting univariate Cox screening, and (iii) pruning highly correlated genes. We then standardized the training set (with the same centering/scaling applied to validation data). These preprocessing steps were not applied in the PCA-based workflow.

Table 20: Table 16: Model Performance Comparision

Model_Type	Mean_Cox_iAUC	Mean_Cox_Cindex	Mean_RSF_iAUC	Mean_RSF_Cindex	Cox_Sum	RSF_Sum	Predictors	Best_Performance	Best_Iteration	Best_Seed	Best_Method	Global_PH_Test_p
Clinical-only	0.5544709	0.5237260	0.5164750	0.5001670	1.078197	1.016642	er, grad....	1.515951	28	92000028	Cox	0.0002183
Genomic-only	0.5212806	0.5161470	0.5199415	0.5076450	1.037428	1.027586	ge_2007...	1.474515	28	92000028	RSF	NA
Clinical-Genomic Integrated	0.5410127	0.5391948	0.5220367	0.5039948	1.080207	1.026031	age, er,....	1.497065	28	92000028	RSF	NA
PCA-Genomic-only	0.5712670	0.5623551	NA	NA	1.133622	NA	NA	1.604300	92	12487	COX	0.0000000
VAE-Genomic-only	0.5427196	0.5391860	NA	NA	1.081906	NA	NA	1.417624	50	[illegible]	COX	0.0000000

Based on the test results, none of the models demonstrate satisfactory discriminative performance on the test set. The average scores are around 0.5, only slightly better than random classification. One reason for this is the small sample size, as well as random factors such as hospital-related effects. In addition, the algorithms and overall approach still require further improvement.

Among all models, the Cox model constructed after applying PCA for dimensionality reduction on the gene expression data performed better on average than the others. In its best iteration, the combined iAUC + C-index reached 1.60, but the proportional hazards (PH) test was not passed. Nevertheless, since the project requires selecting one model as the predictive model for future use, we have chosen this model as the predictive model.

Model application and Prediction

In order to perform personalized survival prediction for new patients using a pre-trained PCA-COX model, the genomic data is needed to be standardized using the same center and scale parameters from the original training set to maintain consistency. Then the standardized data is transformed into principal component space using the rotation matrix from the trained PCA model, retaining only the most significant components based on the original variance threshold.

The transformed data creates the input for COX regression prediction. The pre-trained COX model then generates three key predictions: a linear risk score indicating relative risk, a hazard ratio representing the multiplicative increase in risk compared to the baseline, and most importantly, a personalized survival probability estimate specifically at the patient's own observed follow-up time.

Table 21: PCA Cox prediction VS real survive time

	risk_score	hazard_ratio	survival_P_at_val_time	val_time_dmfs	val_event_dmfs
GSM178024	-0.7873754	0.4550375	0.8302494	6135	0

Discussion

We evaluated multiple Cox proportional hazards (PH) models and random survival forests (RSF) on clinical and gene-expression data, and also built Cox models on PCA and VAE–derived representations. All models performed suboptimally. Likely contributors include: (1) limited sample size and few events—low events-per-variable (EPV) leading to unstable/overfit estimates; (2) violations of the PH assumption; (3) the $p \gg n$ regime and multicollinearity in gene data, which harm stability and generalization; (4) suboptimal hyperparameters (e.g., for VAE and RSF), suggesting a grid search over latent_dim, hidden_dim, ntree, etc.; and (5) weak signal—clinical/genomic predictors may have limited prognostic value, motivating alternative biological stratifications or data types.

For follow-up, one could fit RSF on PCA scores and on VAE latent variables. The relatively better performance of PCA may reflect preprocessing choices—for example, the absence of univariate Cox/LASSO pre-screening and/or omission of scaling before PCA in some configurations. These steps warrant careful reconsideration to avoid degrading signal representation or injecting unnecessary noise.

Conclusion

Genomic data are increasingly important for cancer prognosis. Using gene-expression and clinical data from the GEO database (GSE7390), we conducted a survival analysis of breast cancer and compared Cox proportional hazards (PH) models with Random Survival Forests (RSF) across different feature representations. Missing clinical values were handled via multiple imputation by chained equations (MICE). To address high dimensionality in the genomic data, we explored PCA and VAE for dimensionality reduction, L1-penalized Cox, median absolute deviation (MAD) filtering, and correlation pruning. All preprocessing and feature selection were fit on the training set and then applied to the validation set to prevent data leakage. Model performance was evaluated using the C-index and the integrated time-dependent AUC (iAUC). Training and testing employed repeated bootstrap resampling to ensure robust, less-biased estimates. The model ultimately chosen for prospective use was a Cox PH model built on PCA-derived principal components. Although validation performance did not reach a clearly discriminative level, the study provides a useful baseline and guidance for future refinement.